

Byungdoo Kong

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RESEARCH INTERESTS

Stochastic control and games; mathematical finance; actuarial science.

EDUCATION

- **University of California, Santa Barbara** Sep. 2024 – Present
Ph.D. in Statistics and Applied Probability (Advisor: Prof. Ruimeng Hu) Santa Barbara, CA, USA
- **Sungkyunkwan University** Feb. 2024
M.Sc. in Actuarial Science (Advisor: Prof. Hangsuck Lee) Seoul, Korea
- **Sungkyunkwan University** Feb. 2022
B.Sc. in Mathematics Seoul, Korea

WORKING PAPERS

- [2] **Endogenous risk classification and class-specific reinsurance pricing in competitive insurance markets: Finite-player and mean-field analysis** (with R. Hu). *In preparation.*
- [1] **Endogenous reinsurance pricing in large competitive insurance markets: Finite-player and mean-field analysis** (with R. Hu). *Submitted*, arXiv:2606.27150, 2026.

PUBLICATIONS

- [11] **Analysing reforms of social security disability pension provision in a multistage overlapping generations model** (with H. Lee, S. Haberman, and S. Y. Jeong). *Journal of Risk and Insurance*, forthcoming, 2026.
- [10] **Valuation of multi-step double-barrier options with time-varying rates and jumps: Semi-closed approach** (with H. Lee, M. Lee, and H. Ha). *International Review of Financial Analysis*, 118, 105341, 2026.
- [9] **Valuing American strangle options via double barrier first-touch digitals** (with H. Lee, M. Lee, and H. Ha). *Applied Economics*, 1–21, 2026.
- [8] **A first-touch approach to American option valuation with piecewise linear boundaries** (with H. Lee, M. Lee, and H. Ha). *Finance Research Letters*, 98, 109843, 2026.
- [7] **Age-specific multi-stage OLG model for PAYG pension schemes** (with H. Lee, J. Hong, and S. Y. Jeong). *Computational Economics*, 67(3), 1463–1510, 2026.
- [6] **Two-asset double barrier options** (with H. Lee, H. Ha, and G. Lee). *Computational Economics*, 66(2), 1071–1106, 2025.
- [5] **Multi-step double barrier options under time-varying interest rates** (with H. Lee, Y. Kye, and S. Song). *The North American Journal of Economics and Finance*, 76, 102372, 2025.
- [4] **Foreign equity lookback options with partial monitoring** (with H. Lee and H. Ha). *Finance Research Letters*, 67, 105726, 2024.
- [3] **Valuing three-asset barrier options and autocallable products via exit probabilities of Brownian bridge** (with H. Lee, H. Ha, and M. Lee). *The North American Journal of Economics and Finance*, 73, 102174, 2024.
- [2] **Pricing first-touch digitals with a multi-step double boundary and American barrier options** (with H. Lee and H. Ha). *Finance Research Letters*, 59, 104699, 2024.
- [1] **Pricing multi-step double barrier options by the efficient non-crossing probability** (with H. Lee, H. Ha, and M. Lee). *Finance Research Letters*, 54, 103772, 2023.

PRESENTATIONS

- [6] **Endogenous reinsurance pricing in large competitive insurance markets: Finite-player and mean-field analysis**, UCSB Financial Math Student Seminar (FMSS), Santa Barbara, CA, USA, Oct. 2026.
- [5] **Finite-player and mean-field Stackelberg reinsurance-investment games with relative performance and common noise**, *The 10th Asian Quantitative Finance Conference (AQFC 2026)*, Pohang, Korea, July 2026.
- [4] **Finite-player and mean-field Stackelberg–Nash equilibria in reinsurance-investment games**, *The 29th International Congress on Insurance: Mathematics and Economics (IME 2026)*, Seoul, Korea, June 2026.

- [3] **Optimal early surrender value and surrender penalty rate for variable annuities**, 2023 Winter Conference of Korean Insurance Academic Society, Busan, Korea, Feb. 2024.
- [2] **Multi-step double barrier rebate options**, 2023 Korea Insurance Joint Conference, Cheonan, Korea, Aug. 2023.
- [1] **Valuation of two-asset double barrier products**, 2023 Spring Conference of Korea Risk Management Society, Seoul, Korea, Mar. 2023.

FELLOWSHIPS, GRANTS, AND AWARDS

- **GSA Conference Travel Grant** Sep. 2026
Graduate Student Association, UC Santa Barbara, USA
- **Graduate Student Travel Grant** June 2026
Department of Statistics and Applied Probability, UC Santa Barbara, USA
- **Korea Life Insurance Association Overseas PhD Fellowship** Sep. 2025 – Present
Korea Life Insurance Association, Korea
- **Brain Korea 21 Plus Research Encouragement Scholarship** Mar. 2023
Ministry of Education, Science and Technology, Korea
- **Brain Korea 21 Plus Fellowship** Mar. 2022 – Feb. 2024
Ministry of Education, Science and Technology, Korea
- **Recommendation Scholarship** Sep. 2021
Sungkyunkwan University, Korea
- **Student Success Scholarship** Mar. 2020
Sungkyunkwan University, Korea

TEACHING AND RESEARCH EXPERIENCE

- **University of California, Santa Barbara** Sep. 2024 – Present
Teaching Assistant
 - PSTAT 120A, 120B, and 120C (Probability and Statistics)
 - PSTAT 160A (Applied Stochastic Processes)
 - PSTAT 171 (Mathematics of Fixed Income Markets)
 - PSTAT 172A (Actuarial Statistics); PSTAT 173 (Risk Theory)
- **Korea National Open University** Aug. 2023 – Dec. 2023
Teaching Assistant
 - AI in Finance
- **Institute of Basic Science, Sungkyunkwan University** May 2023 – Aug. 2023
Research Assistant
 - Derived exit-probability formulas for a 3D Brownian bridge.
 - Implemented an algorithm to estimate exit probabilities of a 3D Brownian motion.
- **Sungkyunkwan University** Mar. 2022 – Feb. 2023
Teaching Assistant
 - GEDB001 (Calculus 1); GEDB003 (Linear Algebra)

CERTIFICATIONS

- **Associate of the Society of Actuaries (ASA), in progress**, Society of Actuaries Present
 - Passed Exams and Modules: P, FM, IFM, FAM-L, SRM, STAM, PAF, ASF, FAP.
 - VEEs: Economics; Accounting and Finance; Mathematical Statistics.
- **Financial Risk Manager (FRM) Exam Part I**, Global Association of Risk Professionals May 2022
- **Certified Financial Risk Manager**, Korea Financial Investment Association Sep. 2021
- **Certified Investment Manager**, Korea Financial Investment Association Feb. 2021
- **Advanced Data Analytics Semi-Professional**, Korea Data Agency Dec. 2020

LANGUAGES AND TECHNICAL SKILLS

- **Languages:** Korean (Native), English (Fluent), Chinese (Intermediate).
- **Programming languages:** Python, R, MATLAB, Mathematica, $\text{\LaTeX}$.